from chapter 7 that a *cipher suite* is a collection of cryptographic algorithms chosen to work well together). This guidance is aimed at securing traditional web server applications and their clients, but these cipher suites are not always suitable for IoT use for several reasons:

- The size of code required to implement these suites securely can be quite large and require many cryptographic primitives. For example, the cipher suite `ECDHE-RSA-AES256-SHA384` requires implementing Elliptic Curve Diffie-Hellman (ECDH) key agreement, RSA signatures, AES encryption and decryption operations, and the SHA-384 hash function with HMAC!
- Modern recommendations heavily promote the use of AES in Galois/Counter mode (GCM), because this is extremely fast and secure on modern Intel chips due to hardware acceleration. But it can be hard to implement securely in software on constrained devices and fails catastrophically if misused.
- Some cryptographic algorithms, such as SHA-512 or SHA-384, are rarely hardware accelerated and are designed to perform well when implemented in software on 64-bit architectures. There can be a performance penalty when implementing these algorithms on 32-bit architectures, which are very common in IoT devices. In low-power environments, 8-bit microcontrollers are still commonly used, which makes implementing such algorithms even more challenging.
- Modern recommendations concentrate on cipher suites that provide *forward secrecy* as discussed in chapter 7 (also known as *perfect forward secrecy*).